Q1. Movies released after 2014 SQL Solved

Using hints except Complete Solution is Penalty free now **Use Hint**

Context:

Acme Studios, a major Hollywood production company, maintains a vast movie database. This database tracks detailed information about their films, including titles, directors, and ratings. Understanding audience preferences is crucial for informing creative decisions and attracting viewers.

You are a data analyst at Acme Studios. Your manager has asked you to identify the titles of movies released after 2014 that have received an average vote rating greater than 7. This information can be valuable for Acme Studios to identify movies that resonate with audiences and potentially influence future content creation or marketing strategies.

Problem Statement:

Write a query to display the titles of the movies that are released (i.e., release_year) after **2014** and have an average vote rating (i.e.,vote_average) **greater than 7.**

Result:

- Return the column '**original_title**'.
- Return the result ordered by **original_title** in ascending order.

Dataset description for `movies` table:

1) id - tmdb movie id
2) imdb_id - imdb movie id
3) popularity -A numeric quantity specifying the movie's popularity.
4) budget -The budget in which the movie was made.
5) revenue - The worldwide revenue generated by the movie.
6) original_title- The title of the movie
7) cast - The name of the lead and supporting actors.
8) homepage - A link to the homepage of the movie.
9) director - The name of the director of the movie
10) tagline - Movie's tagline.
11) keywords -The keywords or tags related to the movie.
12) overview -A brief description of the movie.
13) runtime -The running time of the movie in minutes.
14) genres -The genres of the movies
15) production_companies-The production house of the movie.
16) release_date -the date on which it was released.
17) vote_count -the count of votes received.
18) vote_average - average ratings the movie received.
19) release_year - the year on which it was released.

Sample Input:

Table: movies

id	imdb_id	popularity	budget	revenue	original_title	cast	homepage	director	tagline	keywords	overview	runtime	genres	production_companies	release_date	vote_count	vote_average	release_year	budget_adj	revenue_adj
135397	tt0369...	32.9858	15000...	15135...	Jurassic World	Chris...	http://ww...	Colin...	The p...	monster...	Twenty-...	124	Action|...	Universal Studios|Am...	6/9/2012	5562	6.5	2012	139000000	1392450000
262500	tt2908...	13.1125	11000...	29523...	Insurgent	Shail...	http://ww...	Robert...	One C...	based o...	Beatric...	119	Advent...	Summit Entertainment...	3/18/2017	2480	6.3	2017	101200000	271619000
168259	tt2820...	9.33501	19000...	15082...	Furious 7	Vin...	http://ww...	James...	Venge...	car race...	Deckar...	137	Action|...	Universal Pictures|Ot...	4/1/2010	2947	7.3	2010	174800000	1385750000
167073	tt2381111	3.22733	11000...	62076...	Brooklyn	Saoir...	http://ww...	John...	Two c...	love tria...	In 1950...	111	Drama...	Wildgaze Films|Irish Fi...	11/4/2015	754	7.3	2015	10120000	57110000
258480	tt2402...	2.88323	11800...	40272...	Carol	Cate...	http://car...	Todd...	Some...	gay/le...	In 1950...	118	Roman...	Killer Films|Film4|Num...	11/20/2015	562	7.1	2015	10850000	37050000
331781	tt2870...	0.738404	0	8413144	Amy	Amy...	http://ww...	Asif K...	The gi...	jazz/dru...	A docu...	128	Docum...	On The Corner Films|...	7/3/2015	345	7.5	2015	0	7740090

Sample Output:

original_title
Amy
Brooklyn
Carol

```
select original_title
from movies
where (release_year > '2014' and vote_average > '7')
order by original_title asc;
```

Q2. Selected Departments SQL Solved

Using hints except Complete Solution is Penalty free now Use Hint

Problem Statement:
Write a query to find the details of those employees who work in the departments with numbers included in **30**, **40**, or **90**.

- Return the result ordered by **employee_id** in ascending order.

Table: employees

Column Name	Type
employee_id	INT
first_name	VARCHAR
last_name	VARCHAR
email	VARCHAR
phone_number	VARCHAR
hire_date	DATE
job_id	VARCHAR
salary	INT
commission_pct	FLOAT
manager_id	INT
department_id	INT

Sample Input:
Table: employees

employee_id	first_name	last_name	email	phone_number	hire_date	job_id	salary	commission_pct	manager_id	department_id
100	Steven	King	SKING	515.123.4567	1987-06-17	AD_PRES	25000	NULL	NULL	90
116	Shelli	Baida	SBAIDA	515.127.4563	1997-12-24	PU_CLERK	2900	NULL	114	30
120	Matthew	Weiss	MWEISS	650.123.1234	1996-07-18	ST_MAN	8000	NULL	100	50
134	Michael	Rogers	MROGERS	650.127.1834	1998-08-26	ST_CLERK	2900	NULL	122	50
156	Janette	King	JKING	011.44.1345.429268	1996-01-30	SA_REP	10000	0.35	146	80
176	Jonathon	Taylor	JTAYLOR	011.44.1644.429265	1998-03-24	SA_REP	8600	0.2	149	80
199	Douglas	Grant	DGRANT	650.507.9844	2000-01-13	SH_CLERK	2600	NULL	124	50
203	Susan	Mavris	SMAVRIS	515.123.7777	1994-06-07	HR_REP	6500	NULL	101	40

Sample output:

employee_id	first_name	job_id	department_id
100	Steven	AD_PRES	90
116	Shelli	PU_CLERK	30
203	Susan	HR_REP	40

Explanation: The above employees have department_id included in 30, 40, or 90.

```
select employee_id,first_name,job_id, department_id
from employees
where department_id in(30,40,90)
order by employee_id;
```

Q3. Office Codes SQL Solved

Using hints except Complete Solution is Penalty free now Use Hint

Problem Statement:

Write a query to find the employees whose officeCodes are not included in **4,6,7**.

- Return the result ordered by **firstName** and **lastName** in ascending order.

Table: employees

Column Name	Type
employeeNumber	INT
lastName	VARCHAR
firstName	VARCHAR
extension	VARCHAR
email	VARCHAR
officeCode	INT
reportsTo	VARCHAR
jobTitle	VARCHAR

Sample Input:

Table: employees

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1002	Murphy	Diane	x5800	dmurphy@classicmodelcars.com	1	NULL	President
1056	Patterson	Mary	x4611	mpatterso@classicmodelcars.com	1	1002	VP Sales
1088	Patterson	William	x4871	wpatterson@classicmodelcars.com	6	1056	Sales Manager (APAC)
1102	Bondur	Gerard	x5408	gbondur@classicmodelcars.com	4	1056	Sale Manager (EMEA)
1188	Firrelli	Julie	x2173	jfirrelli@classicmodelcars.com	2	1143	Sales Rep
1286	Tseng	Foon Yue	x2248	ftseng@classicmodelcars.com	3	1143	Sales Rep
1323	Vanauf	George	x4102	gvanauf@classicmodelcars.com	3	1143	Sales Rep
1370	Hernandez	Gerard	x2028	ghernande@classicmodelcars.com	4	1102	Sales Rep
1501	Bott	Larry	x2311	lbott@classicmodelcars.com	7	1102	Sales Rep
1702	Gerard	Martin	x2312	mgerard@classicmodelcars.com	4	1102	Sales Rep

Sample output:

firstName	lastName	jobTitle
Diane	Murphy	President
Foon Yue	Tseng	Sales Rep
George	Vanauf	Sales Rep
Julie	Firrelli	Sales Rep
Mary	Patterson	VP Sales

Explanation: The above table shows the employees whose office codes are not 4,6 or 7.

```
select firstName,lastName,jobTitle
from employees
where officeCode not in(4,6,7)
order by firstName,lastName;
```

Q4. Horror genre SQL Solved

Using hints except Complete Solution is Penalty free now **Use Hint**

Problem Statement:

Display the details of the movies which belong to the '**Horror**' genre.

- Return the columns '**original_title**', and '**popularity**'.
- Return the result ordered by the **popularity** in descending order.

Dataset description for `movies` table:

1) id - tmdb movie id
2) imdb_id - imdb movie id
3) popularity -A numeric quantity specifying the movie's popularity.
4) budget -The budget in which the movie was made.
5) revenue - The worldwide revenue generated by the movie.
6) original_title- The title of the movie
7) cast - The name of the lead and supporting actors.
8) homepage - A link to the homepage of the movie.
9) director - The name of the director of the movie
10) tagline - Movie's tagline.
11) keywords -The keywords or tags related to the movie.
12) overview -A brief description of the movie.
13) runtime -The running time of the movie in minutes.
14) genres -The genres of the movies
15) production_companies-The production house of the movie.
16) release_date -the date on which it was released.
17) vote_count -the count of votes received.
18) vote_average - average ratings the movie received.
19) release_year - the year on which it was released.

Sample Input:

Table: movies

id	imdb_id	popularity	budget	revenue	original_title	cast	homepage	director	tagline	keywords	overview	runtime	genres	production_companies	release_date	vote_count	vote_average	release_year	budget_adj	revenue_adj
135397	tt0369...	32.6858	15000...	1615...	Jurassic World	Chris...	http://ww...	Colin...	The p...	monster...	Twenty-...	124	Action\Adventure\...	Universal Studios\Am...	6/9/2012	5562	6.5	2012	138000000	1363528000
76341	tt1392...	28.4199	15000...	37843...	Mad Max: Fury Road	Tom...	http://ww...	Georg...	What...	future\c...	An apo...	120	Action\Adventure\...	Village Roadshow Pict...	5/13/2015	6185	7.1	2015	137000000	3481613000
262500	tt2908...	13.1125	11000...	26523...	Insurgent	Shaile...	http://ww...	Robert...	One C...	based o...	Beatric...	119	Adventure\Science...	Summit Entertainment...	3/18/2015	2480	6.3	2015	110000000	271619000
264660	tt2470...	6.11885	15000...	36889...	Ex Machina	Domh...	http://ww...	Alex...	There...	dancing...	Caleb,...	108	Horror	DNA Films\Universal...	1/21/2015	2854	7.6	2015	13800000	33919800
260346	tt2446...	5.74976	48000...	32577...	Taken 3	Liam...	http://ww...	Olivier...	It End...	revenge...	Ex-gov...	109	Horror	Twentieth Century Fox...	1/1/2015	1578	6.1	2015	44160000	299710000
216015	tt2322...	4.7104	40000...	96985...	Fifty Shades of Grey	Dakot...	https://w...	Sam T...	Are yo...	based o...	When c...	125	Horror	Focus Features\Trigge...	2/11/2015	1885	5.3	2015	36800000	524079000

Sample Output:

original_title	popularity
Ex Machina	6.11885
Taken 3	5.74976
Fifty Shades of Grey	4.7104

```
select original_title,popularity
from movies
where genres = 'Horror'
order by popularity desc;
```

Q5. Recyclable and Low Fat Products SQL Solved

Using hints except Complete Solution is Penalty free now Use Hint

Table: Products

Column Name	Type
product_id	int
low_fats	enum
recyclable	enum

product_id is the primary key for this table.
low_fats is an ENUM of type ('Y', 'N') where 'Y' means this product is low fat and 'N' means it is not.
recyclable is an ENUM of types ('Y', 'N') where 'Y' means this product is recyclable and 'N' means it is not.

Write an SQL query to find the ids of products that are both low fat and recyclable.
Return the result table sorted by product_id in ascending order
The query result format is in the following example.

Example :

Table: Products

product_id	low_fats	recyclable
0	Y	N
1	Y	Y
2	N	Y
3	Y	Y
4	N	N

Output:

product_id
1
3

```
select product_id
from Products
where (low_fats = 'Y' and recyclable = 'Y')
order by product_id;
```

Q6. Low-Quality Problems SQL Solved

Using hints except Complete Solution is Penalty free now Use Hint

Table: Problems

Column Name	Type
problem_id	int
likes	int
dislikes	int

problem_id is the primary key column for this table.
Each row of this table indicates the number of likes and dislikes for a LeetCode problem.
Write an SQL query to report the IDs of the low-quality problems. A LeetCode problem is low-quality if the like percentage of the problem (number of likes divided by the total number of votes) is strictly less than 60%.
Return the result table ordered by problem_id in ascending order.
The query result format is in the following example.

Example :

Problems table:

problem_id	likes	dislikes
6	1290	425
11	2677	8659
1	4446	2760
7	8569	6086
13	2050	4164
10	9002	7446

Output:

problem_id
7
10
11
13

```
select problem_id
from Problems
where (likes/(likes+dislikes) < 0.60)
order by problem_id;
```

Q7. Filter Movies SQL Solved

Using hints except Complete Solution is Penalty free now Use Hint

Consider a database schema for movies with the following table:
Movies: movie_id (Integer, Primary Key), movie_title (Text), genre (Text), release_year (Integer)
Write an SQL query to retrieve the titles of movies **released in 2000** or later that belong to the genres **"Sci-Fi" or "Action."**
Return the result ordered by movie_id in ascending order.

Sample Input:

Table: Movies

movie_id	movie_title	genre	release_year
1	Inception	Sci-Fi	2010
2	Titanic	Romance	1997
3	The Dark Knight	Action	2008
4	Forrest Gump	Drama	1994
5	Avatar	Sci-Fi	1999

Sample Output:

movie_title
Inception
The Dark Knight

```
select movie_title
from Movies
where (release_year >= '2000' and genre in('Sci-Fi','Action'))
order by movie_id;
```

Q8. Filter Customers SQL Solved

Using hints except Complete Solution is Penalty free now Use Hint

Consider a database schema for customers with the following table:
Customers: customer_id (Integer, Primary Key), customer_name (Text), last_purchase_date (Date)
Write an SQL query to find the names of customers who have **not** made a purchase since **January 1, 2023**.
Return the results ordered by customer_id in ascending order.

Sample Input:

Table: Customers

customer_id	customer_name	last_purchase_date
1	Alice	2022-10-15
2	Bob	2023-01-20
3	Carol	2023-03-05
4	David	2022-12-01
5	Eve	2023-02-10

Sample Output:

customer_name
Alice
David

```
select customer_name
from Customers
where last_purchase_date < '2023-01-01'
order by customer_id asc;
```

Q9. Filter Orders SQL Solved

Using hints except Complete Solution is Penalty free now Use Hint

Consider a database schema for orders with the following table:
Orders: order_id (Integer, Primary Key), customer_id (Integer), order_status (Text)
Write an SQL query to retrieve the order IDs of orders placed by **customer 1** that are either **"Shipped" or "Delivered."**
Return the output ordered by order_id in ascending order.

Sample Input:

Table: Orders

order_id	customer_id	order_status
1	1	Shipped
2	2	Shipped
3	1	Pending
4	1	Delivered
5	3	Pending

Sample Output:

order_id
1
4

```
select order_id
from Orders
where customer_id = '1' and order_status in('Shipped','Delivered')
order by order_id;
```

Q10. Filter Movies not in Specific Genre SQL Solved

Using hints except Complete Solution is Penalty free now Use Hint

Consider a database schema for movies with the following table:
Movies: movie_id (Integer, Primary Key), movie_title (Text), genre (Text)
Write an SQL query to retrieve the titles of movies that are **not in** the genres **"Romance" or "Drama."**
Return the output ordered by movie_id in ascending order.

Sample Input:

Table: Movies

movie_id	movie_title	genre	release_year
1	Inception	Sci-Fi	2010
2	Titanic	Romance	1997
3	The Dark Knight	Action	2008
4	Forrest Gump	Drama	1994
5	Avatar	Sci-Fi	1999

Sample Output:

movie_title
Inception
The Dark Knight
Avatar

```
select movie_title
from Movies
where genre not in('Romance','Drama')
order by movie_id;
```